graded_bets_max

equity strategy | family: equity-max | status: research | derived from `graded_bets`

What this strategy does

graded_bets with bigger bets for more return: base unit 0.005, 6x leverage cap

Why it should work. Same drift as Extreme Picks but harvested across the whole cross-section for lower variance.

Design notes (from the strategy's source file).

Max-size variants ("<name>_max") of the hand-built equity strategies.

Same signal, trim, tilt and neutralization as each base strategy; only the two dials that set how big and how levered the book is change (base_unit_fraction, gross-leverage cap = 6x). The multiplier for each was picked by sweeping 1.5x..12x the original base unit and taking the highest return subject to Sharpe >= 0.6 and max drawdown >= -40% (a worse drawdown is a fund-ending event, not a risk preference). The untouched 5%-of-ADV liquidity cap remains the real capacity constraint, and bites roughly four times as often at this sizing.

These are generated from the base strategy classes, so a change to a base strategy's weights flows through automatically.

parameter	value
base_unit_fraction	0.005
max_gross_leverage	6.0
liquidity_cap_frac	0.05
return_basis	mktadj
cap_mode	proportional
exposure_cap	open_book
beta_overlay	0.0

Performance

metric	graded bets max	market (CRSP VW)
Annualized return	8.61%	9.94%
Annualized volatility	14.43%	19.35%
Sharpe (NAV-pct)	0.64	0.59
Sortino	0.91	
Max drawdown	-48.0%	-55.5%
Calmar	0.18	
Final NAV ($10M start)	$49,935,311	
Worst day	-7.90%	
% positive days	50.0%	

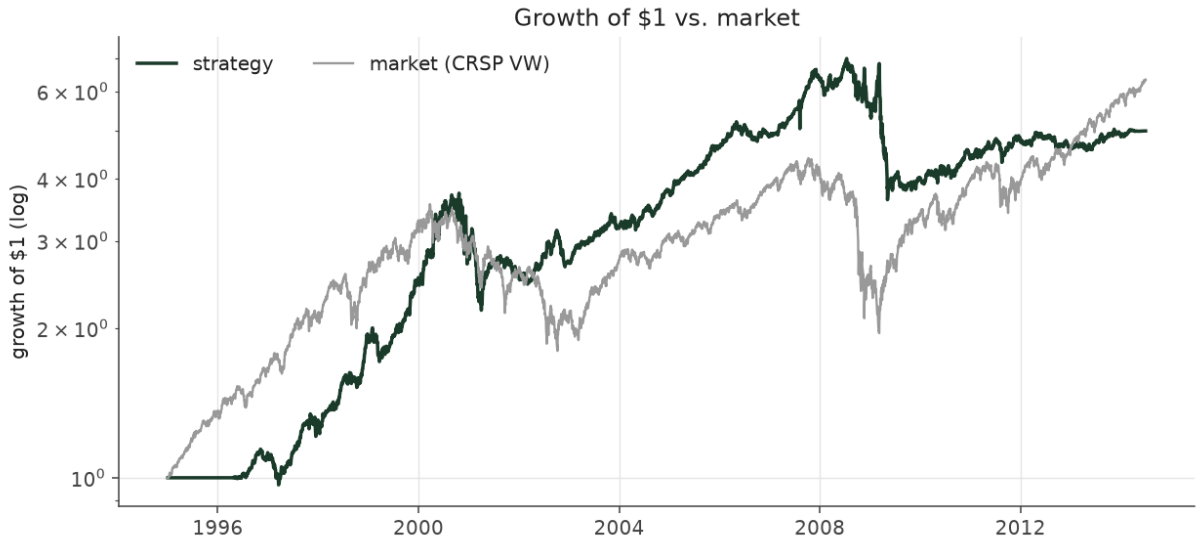
Sharpe 1996-2001	0.97	
Sharpe 2002-2007	1.53	
Sharpe 2008-2013	-0.20	

Correlation of daily returns with the market: **0.02**.

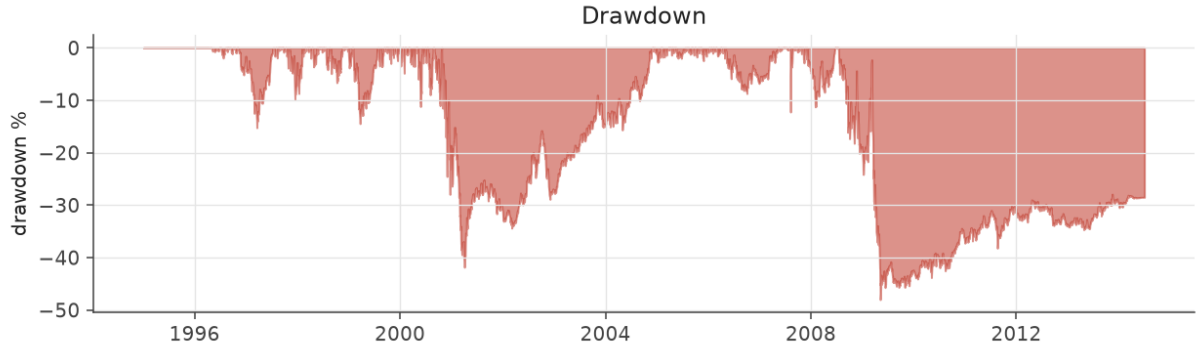
• best Sharpe among equity: `calm_rank_tilt` (1.81); this one: 0.645

• best return among equity: `market_boost_max` (0.15); this one: 0.0861

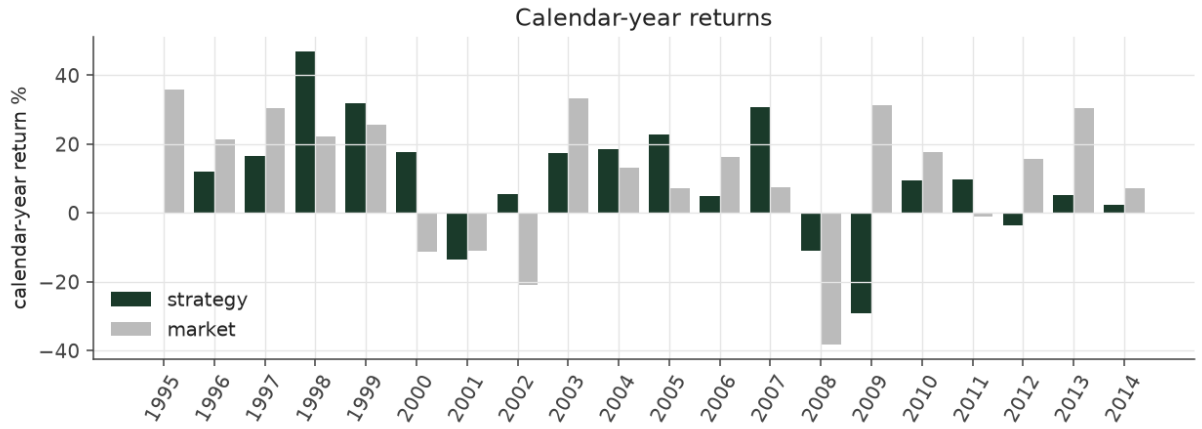
• best Calmar among equity: `calm_rank_tilt` (0.614); this one: 0.179



NAV (log scale)



Drawdown from running peak

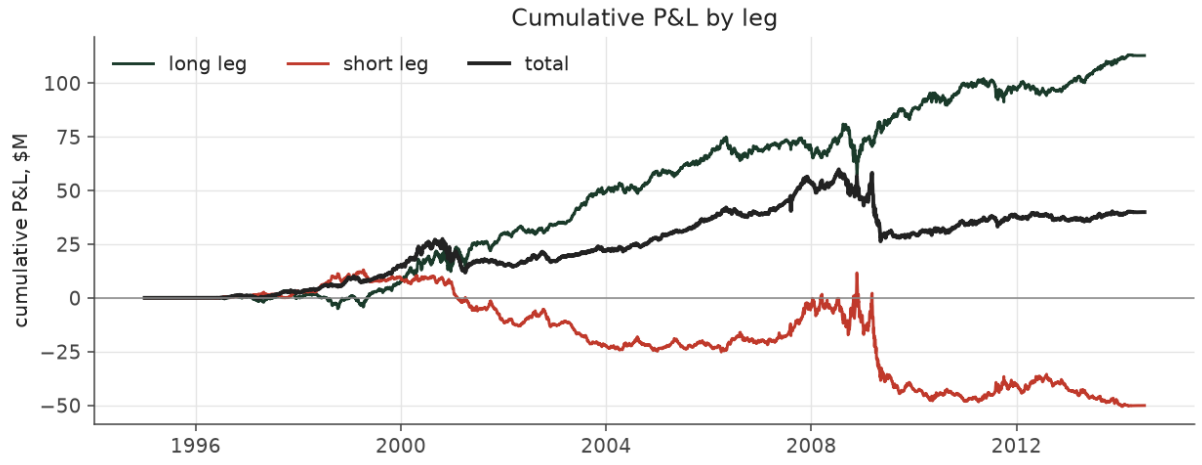


Calendar-year returns



Is the edge stable, or concentrated in a few years?

Where the return comes from



Long leg vs. short leg vs. total

	long leg	short leg
P&L ($M, before costs)	112.84	-50.12
P&L after costs ($M)	101.20	-61.26
of which common (average-event) component ($M)	100.50	-100.16

of which excess / PEAD component ($M)	12.34	50.05
excess minus costs ($M)	0.70	38.90
gross return per $ traded (bp)	165.63	-84.27
positions in the money (hit rate)	0.55	0.48
stand-alone Sharpe (leg only)	0.90	-0.30

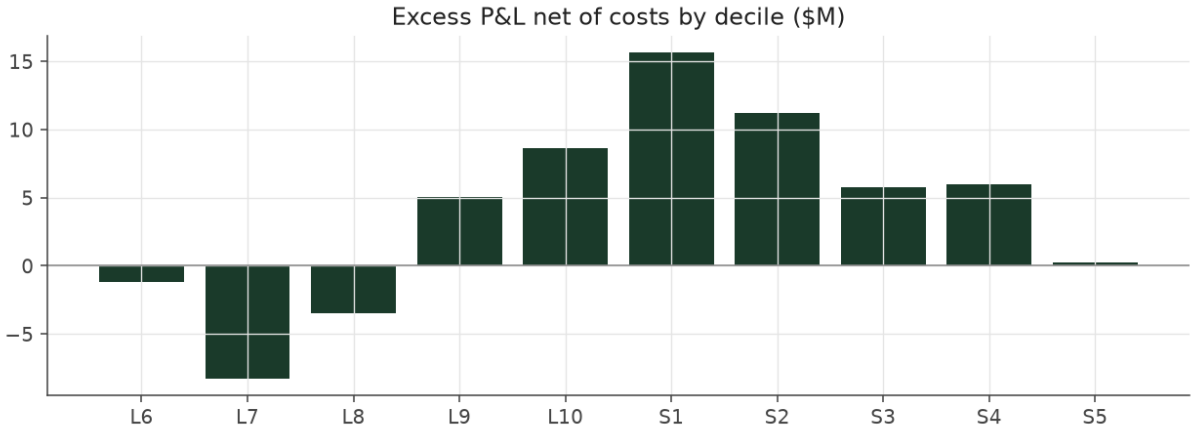
Daily P&L correlation between the two legs: **-0.70**.

• **The short leg loses money in absolute terms** (-50.1M before costs) while the long leg earns +112.8M. That is mostly a benchmark effect, not a signal failure: the average earnings-event stock beat the cap-weighted market during its holding window, which flatters the long leg and hurts the short leg by 100.5M / 100.2M. Stripping that common component, the surprise signal itself earned +12.3M on the long side and +50.0M on the short side.

• The two legs' daily P&L are -0.70 correlated: the short leg is the hedge that makes the long leg's variance disappear (stand-alone leg Sharpe 0.90 long / -0.30 short vs 0.64 combined).

By decile and side (signal-driven part only)

leg	decile	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
long	6	12,092	189.84	1.01	-0.77	0.39	-1.17	-2.9%	-61.47
long	7	20,923	873.44	6.24	-6.71	1.59	-8.30	-21.0%	-95.02
long	8	16,617	1,290.56	16.87	-1.23	2.25	-3.48	-8.8%	-26.96
long	9	18,570	1,870.55	30.29	8.15	3.12	5.03	12.7%	26.89
long	10	20,136	2,588.20	46.79	12.91	4.29	8.62	21.8%	33.29
short	1	18,299	2,067.84	-21.86	19.51	3.85	15.65	39.5%	75.71
short	2	18,907	1,786.86	-19.20	14.55	3.30	11.25	28.4%	62.96
short	3	15,390	1,149.57	-11.73	7.90	2.16	5.74	14.5%	49.91
short	4	20,837	829.28	-7.02	7.62	1.61	6.01	15.2%	72.53
short	5	11,143	113.75	-1.45	0.47	0.23	0.25	0.6%	21.64



D10 and the short deciles carry the alpha; the middle of each leg earns little

By size quintile (1 = smallest)

size quintile	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $

1	19,552	593.07	-5.07	-0.79	2.11	-2.90	-7.3%	-48.87
2	32,133	2,042.29	10.76	18.51	5.35	13.16	33.2%	64.43
3	36,789	3,060.07	15.71	21.98	6.11	15.87	40.1%	51.85
4	40,259	3,476.08	9.04	12.45	5.29	7.17	18.1%	20.62
5	44,181	3,588.39	9.49	10.24	3.93	6.31	15.9%	17.58

By liquidity quintile (1 = least liquid)

liquidity quintile	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
1	19,144	359.93	2.69	5.19	1.80	3.39	8.6%	94.25
2	32,282	1,932.64	5.36	13.93	5.80	8.13	20.5%	42.09
3	37,327	3,303.34	17.47	26.73	6.61	20.13	50.8%	60.93
4	40,479	3,540.52	4.53	8.63	4.96	3.68	9.3%	10.39
5	43,677	3,622.95	9.88	7.87	3.62	4.25	10.7%	11.72

By holding-window length

hold bucket	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
<=40d	68,512	4,427.48	-3.63	24.17	7.85	16.33	41.2%	36.88
41-60d	40,665	3,072.43	5.51	18.75	5.94	12.81	32.3%	41.69
61-90d	47,915	4,396.80	24.44	6.60	6.97	-0.37	-0.9%	-0.84
>90d	15,822	863.18	13.61	12.87	2.03	10.84	27.4%	125.54

By era

era	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
1996-01	49,863	2,068.95	15.13	15.75	3.63	12.12	30.6%	58.56
2002-07	58,152	4,740.85	35.95	39.69	8.29	31.41	79.3%	66.25
2008-13	64,899	5,950.10	-11.14	6.95	10.87	-3.92	-9.9%	-6.58

By FF12 sector

ff12 sector	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
BusEq	32,014	2,290.43	6.72	0.71	4.06	-3.35	-8.4%	-14.60
Chems	3,914	295.35	0.95	1.92	0.50	1.43	3.6%	48.23
Durbl	4,045	293.61	0.09	1.39	0.53	0.86	2.2%	29.35
Enrgy	9,090	671.43	5.88	8.21	1.10	7.11	17.9%	105.84
Hlth	17,523	1,182.32	7.34	6.45	2.24	4.20	10.6%	35.54
Manuf	16,998	1,274.80	-0.24	3.80	2.23	1.56	3.9%	12.25
Money	25,762	2,020.89	3.75	14.66	3.60	11.06	27.9%	54.73
NoDur	7,470	554.15	2.01	2.50	0.95	1.55	3.9%	27.90

Other	26,552	1,952.36	8.79	13.33	3.74	9.59	24.2%	49.12
Shops	17,482	1,282.81	5.79	6.95	2.28	4.67	11.8%	36.38
Telcm	5,624	409.78	-1.22	0.63	0.69	-0.06	-0.2%	-1.48
Utils	6,440	531.98	0.08	1.84	0.85	0.99	2.5%	18.63

By idiosyncratic-volatility tercile

vol tercile	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
high vol	56,962	4,468.58	-3.23	7.19	7.29	-0.09	-0.2%	-0.21
low vol	58,063	4,055.47	23.12	29.47	7.89	21.58	54.5%	53.20
mid vol	55,419	4,097.00	19.68	25.50	7.32	18.18	45.9%	44.37

Short-borrow sensitivity (not modeled in the backtest)

borrow rate	fee M	ann. return	drag
0.0%	0.0	8.61%	-0.00%
0.5%	6.0	7.90%	0.71%
1.0%	12.0	7.09%	1.52%
2.0%	24.0	5.02%	3.58%
5.0%	59.9	-100.00%	108.61%

The engine charges the same transaction cost on shorts as on longs and no stock-borrow fee. The short leg is concentrated in small, illiquid names, where borrow is scarce and expensive -- read the table as a floor on the cost of running the short leg.